

A Markov Prediction Model for Sea-Level Fluctuations Based on Wave Spectrum

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Abstract: As global attention to ocean resources intensifies and the activities of ocean resource development deepen, accurately predicting sea-level fluctuations becomes crucial for ensuring the safety and efficiency of ocean exploration. This study aims to enhance the safety and efficiency of maritime operations through an in-depth analysis of sea-level fluctuations. The study employs wave spectrum for numerical simulation to analyze the energy distribution of waves at different frequencies and uses linear superposition of cosine waves to simulate changes in sea-level fluctuations. Subsequently, based on the simulation results, a Markov model's state transition probability matrix is constructed to predict changes in sea-level fluctuations. This research achieves short-term predictions of sea-level fluctuations. In the final part of the article, measures that should be taken to improve prediction accuracy are discussed, such as applying deep learning techniques and considering the coupling effects among wind, waves, and currents.

Key Words: Wave Spectrum, Numerical Simulation, Markov Chain, Wave Prediction

1 Introduction

As global exploitation of marine resources intensifies, the safety and efficiency of marine resource exploration have become crucial topics in research and practice. Accurately predicting sea-level fluctuations is essential for ensuring the safety of operations [1]. Sea-level fluctuations, one of the most prominent dynamic phenomena in the marine environment, pose significant challenges to precise predictions due to their irregularity and complexity.

The rapid development of computer technology has made it possible to obtain marine data through simulations, providing a means to conduct simulation studies on engineering problems in actual sea conditions. Continuous in-depth research on sea waves has led scholars to propose various wave spectrum theories, such as the P-M spectrum [2] and the JONSWAP spectrum [3], which detail the energy distribution under different sea conditions. For instance, Lv and colleagues conducted a numerical simulation of the waves in the Northwest Pacific using the SWAN wave numerical model, optimizing the simulation parameters [4]. Xia analyzed the hydrodynamic characteristics of seawall revetments using solitary waves as a substitute for tsunami waves [5]. Miles used numerical simulation methods to study the circulation on the continental shelf during Hurricane Sandy's landfall to gain a deeper understanding of the coastal ocean's response to tropical cyclones [6].

Predicting the significant wave height is a crucial factor in ensuring the safety and efficiency of marine resource exploration, and deep learning models, with their excel-

lent adaptive ability and non-linear processing mechanism, have made progress in solving the problem of wave height prediction. For example, Zhang improved prediction accuracy significantly by combining convolutional neural networks and short-term memory networks for wave height prediction [7]. Karan demonstrated high accuracy of the model by first processing data with ensemble empirical mode decomposition and then predicting wave height using bidirectional Long Short-Term Memory (LSTM) networks [8]. Mo's hybrid model, combining ensemble empirical mode decomposition and generative adversarial networks, successfully achieved accurate predictions of wave heights [9].

Despite the progress made by deep learning prediction models, challenges still need to be solved in predicting short-term fluctuations, and there is an urgent need for more efficient wave prediction models. As a tool for describing system state transitions, the Markov model can quickly predict the system's future state through the transition probability matrix. It is particularly suitable for situations requiring rapid acquisition of sea-level state characteristics.

Xia's location prediction method, combining variable-order Markov models and user spatiotemporal rules, improved the accuracy of location predictions [10]. Zaky's stochastic Markov state-switching model, suitable for estimating car drivers' characteristics and extracting different driving state features, can be applied to collision detection, driver assistance, and assessment systems, among others [11]. Pu used an improved Markov chain to predict the first-place result in the men's 100m, achieving high-precision prediction [12].

As a statistical model based on state transitions, the Markov model has been widely researched and applied

This work is supported by the National Natural Science Foundation of China under Grant 62333019, the 111 project under Grant B17040.

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for its ability to handle stochastic processes and predict state changes in the short term. Due to the lack of actual sea-level fluctuation data in this study, this paper initially conducted simulations based on wave spectra, thus generating a substantial dataset for further analysis. Subsequently, this paper applies Markov theory to predict sea-level fluctuations, intending to improve the accuracy of short-term sea-level change predictions to satisfy the requirements for stability control of marine exploration vessels.

2 Sea-Level Fluctuations Datasets

This section elaborates on the fundamental principles of simulating sea-level fluctuations using wave spectra to generate sea-level fluctuations datasets. Initially, it delves into wave theory and the concept of wave spectra, calculating the energy in various frequency bands using the energy density formula. Based on these theories, the mathematical equation for sea-level fluctuation is derived. By applying the wave spectrum formula under specific sea conditions, practical simulation experiments of sea-level fluctuations can be conducted.

In reality, ocean waves exhibit irregular fluctuations. This study assumes that such irregular waves are formed by the superposition of multiple small-amplitude regular waves, where the amplitude of each small wave is much smaller than its wavelength or water depth, allowing these irregular waves to be treated through linear methods:

$$\eta(t) = \sum_{n=1}^{\infty} a_n \cos(\omega_n t + \phi_n) \quad (1)$$

where a_n is the amplitude of the individual small-amplitude wave, ω_n is the angular frequency of the wave, and ϕ_n is the initial phase of the wave. Since irregular waves do not have a fixed wave height, period, or phase, the spectral method is commonly used to simulate the continuous sea-level. The spectral energy density is defined as:

$$S(\omega) = \frac{1}{\Delta\omega} \sum_{\omega}^{\omega+\Delta\omega} \frac{1}{2} a_i^2 \quad (2)$$

where $S(\omega)$ represents the spectral energy density; ω is the angular frequency of each cosine wave composing the sea wave; a_i is the amplitude of the cosine wave corresponding to frequency ω . It is assumed that the energy of the sea wave spectrum is primarily concentrated in the frequency range from ω_L to ω_H , and the energy outside this range is disregarded. Thus, irregular waves can be considered as a linear combination of regular waves within different frequency intervals, and their wave surface equation is defined accordingly:

$$\eta(t) = \sum_{n=1}^M \sqrt{2S(\hat{\omega}_i)\Delta\omega} \cos(\hat{\omega}_i t + \phi_i) \quad (3)$$

The JONSWAP spectrum, derived from wave surface data measured in the "Joint North Sea Wave Project," is a typical wave spectrum applicable to describing wind waves under moderate wind conditions with limited fetch and highly agrees with measured data. For ease of engineering application, Goda proposed the following improved JONSWAP-type spectrum in 1999:

$$S(f) = \beta_j H_s^2 T_p^{-4} f^{-5} \exp \left[-\frac{5}{4} (T_p f)^{-4} \right] \quad (4)$$

where

$$\beta_j = \frac{0.06238}{0.230 + 0.0336\gamma - 0.185(1.9 + \gamma)^{-1} [1.094 - 0.01915 \ln \gamma]} \quad (5)$$

$$T_p = \frac{T_s}{1 - 0.0132(\gamma + 0.2)^{-0.559}} \quad (6)$$

where H_s is the significant wave height; T_s is the significant wave period; T_p is the peak spectrum period; f is the frequency; γ is the peak sharpness factor, with the peak sharpness factor chosen as 3.3 usually. This spectrum form has a clear advantage because once the value of γ is determined, the spectrum shape can be established based on the elements of the design wave.

3 Sea-Level Fluctuations Prediction Model

The Markov chain is a mathematical model that describes the probability of state transitions over time, with its most significant characteristic being "memorylessness." This means the future state of a system is solely determined by its current state, unrelated to previous states. This characteristic significantly simplifies the analysis of complex systems, making Markov chains widely used across various fields. In analyzing sea-level fluctuations, the sea-level can be divided into several states based on different heights, and the fluctuations at a given location can be viewed as transitions between these states:

State a_1 : $0 < h_j < H_c$;

State a_2 : $H_c < h_j < 2H_c$;

...

State a_i : $(i-1)H_c < h_j < iH_c$;

where $j = 1, 2, 3, \dots$, with H_c representing the interval between sea-level heights. Assuming the sea-level state at time t_0 is known, the first-order Markov chains can predict the sea-level state at any future time $t > t_0$, with the prediction only related to the state at t_0 .

The transitions between states are facilitated by a pre-defined transition probability [13]:

$$P(H_{n+k} = a_{n+k} | H_n = a_n, H_{n-1} = a_{n-1}, H_{n-2} = a_{n-2}, \dots, H_1 = a_1) = P(H_{n+k} = a_{n+k} | H_n = a_n) \quad (7)$$

where the transition probability $P(H_{n+k} = a_j | H_n = a_i)$ indicates the probability of the sea-level transition-

ing from state i to state j in k steps. Specifically, when $k = 1$, it is referred to as a first-order Markov chain, where the one-step transition probability accurately describes the immediate transitions between states:

$$p_{ij} = P(H_{n+1} = a_j | H_n = a_i) \quad (8)$$

This one-step transition probability facilitates the construction of a one-step transition probability matrix:

$$p^{(1)} = (p_{ij}) = \begin{pmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mn} \end{pmatrix} \quad (9)$$

where $0 \leq p_{ij} \leq 1$, representing the transition probability, and $\sum_{j=1}^m p_{ij} = 1$.

The conditional probability in the above formula is [14]:

$$p_{ij} = \frac{\int_{(j-1)H_c}^{jH_c} \int_{(i-1)H_c}^{iH_c} P(H_1, H_2) dH_1 dH_2}{\int_{(i-1)H_c}^{iH_c} Q(H_1) dH_1} \quad (10)$$

where $P(H_1, H_2)$ is the joint probability density function of adjacent wave heights H_1 and H_2 [14]:

$$P(H_1, H_2) = \frac{4H_1 H_2}{(1 - 4\rho^2) H_{rms}^4} \cdot I_0 \left(\frac{4H_1 H_2 \rho}{(1 - 4\rho^2) H_{rms}^2} \right) \cdot \exp \left[-\frac{H_1^2 + H_2^2}{H_{rms}^2 (1 - 4\rho^2)} \right] \quad (11)$$

$Q(H_1)$ represents the wave height distribution, following the Rayleigh distribution:

$$Q(H_1) = \frac{H_1}{H_{rms}^2} \exp\left(-\frac{H_1^2}{H_{rms}^2}\right) \quad (12)$$

where I_0 is the modified zeroth-order Bessel function of the first kind, ρ is the correlation parameter between H_1 and H_2 , and H_{rms} is the root mean square wave height. ρ is a function of the correlation coefficient γ_{HH} between H_1 and H_2 :

$$\gamma_{HH} = [E(2\rho) - (1 - 4\rho^2)K(2\rho)/2\pi/4]/(1 - \pi/4) \quad (13)$$

where K and E are the complete elliptic integrals of the first and second kinds, respectively. A higher correlation coefficient γ_{HH} suggests a higher likelihood of consecutive large waves. For instance, γ_{HH} is about 0.2 for wind waves, while it can reach 0.6 for swells from distant oceans.

In simulations using the wave spectrum:

$$H_{rms}^2 = 8m_0 \quad (14)$$

$$m_0 = \int_0^\infty S(\omega) d\omega \quad (15)$$

where m_0 is the zeroth moment of the wave spectrum,

$S(\omega)$ represents the wave spectrum.

Methods to predict the system's next state include the direct probability method and stochastic simulation. The direct probability method selects the next state based on the highest probability of the current state, neglecting the randomness inherent in Markov chains. In contrast, the stochastic simulation method more accurately reflects the system's randomness by generating a random number between 0 and 1 and selecting the next state based on the position of this random number in the cumulative distribution.

4 Prediction Results

In this section, we conduct simulation experiments based on the theories above. Initially, we simulate sea-level fluctuations based on wave spectrum and wave theory to generate the necessary dataset for prediction. Subsequently, by establishing a state transition probability matrix and utilizing Markov chain theory, we predict sea-level fluctuations, eventually obtaining and analyzing the prediction results.

4.1 Sea-Level Fluctuations Simulation

In this study, we opt for the JONSWAP spectrum as our target spectrum, assuming an effective wave height H_s of 4 meters and a peak period T_p of 10 seconds. As illustrated in Fig. 1, the energy distribution is primarily concentrated within the frequency interval from f_L to f_H , with the energy outside this range is negligible. Typically, f_H is set to 3 to 4 times the peak frequency, which is determined to be 0.1 Hz, thus setting the simulated frequency range to 0.07 Hz to 0.4 Hz.

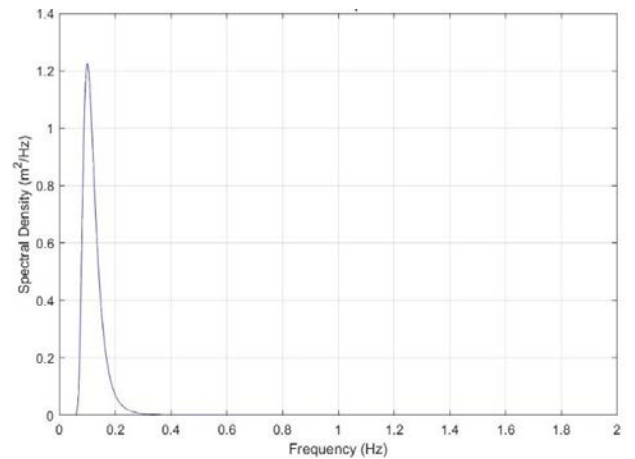


Fig. 1. JONSWAP spectrum

When employing an equal frequency division approach, the number of divisions, M , must be sufficiently large to ensure an acceptable frequency interval. Only with a sufficiently large M can the frequency intervals be small enough to prevent distortion of the simulated wave spectrum near the peak frequency due to sampling frequency that is too low. Furthermore, to ensure the randomness of the simulated waves, the angular frequencies

of the constituent cosine waves are generated using a random function to avoid periodic phenomena in the waves:

$$\omega_i = \hat{\omega} - \frac{1}{2}\Delta\omega + RAN\Delta\omega \quad (16)$$

where RAN is a random number within the $(0,1)$ interval. The initial phase is randomly chosen between 0 and 2π . According to the Shannon sampling theorem, the sampling frequency should be at least twice the highest frequency of the sampled signal. Therefore, the simulation step size dt should be greater than $\frac{1}{2f_H}$, with this study adopting a step size of 0.5 seconds. The simulation parameters are shown in Table 1.

Table 1. Simulation Parameters

H_s	4m
T_p	10s
f_L	0.07Hz
f_H	0.4Hz
γ	3.3
Δt	0.5s

As is shown in Fig.2, the results show the sea-level fluctuations over a total time of 50 seconds, with a time interval of 0.5 seconds, effectively discretizing the continuous wave motion. As shown in Fig. 2, the wave surface height fluctuates significantly within the time frame, exhibiting irregular fluctuation characteristics. The waves also have strong nonlinearity and randomness, and the fluctuation amplitude has no apparent regularity. This irregular fluctuation is typical in predicting and analyzing the marine environment.

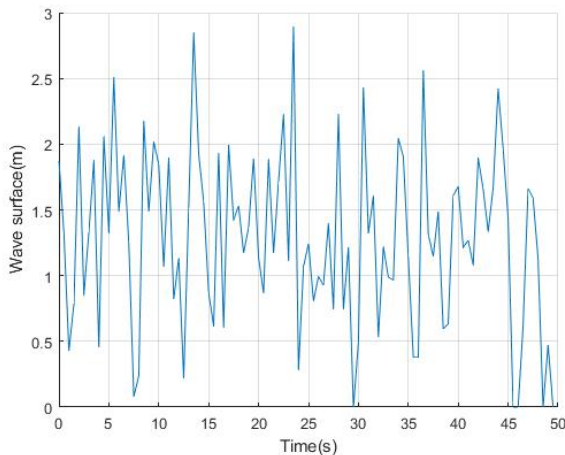


Fig. 2. 50s Numerical Simulation of Sea-Level

4.2 Sea-Level Fluctuations Prediction

In the wave prediction section, considering the focus of this study is on the wave disturbances encountered by exploration vessels during operations, a short prediction time window of 5s was chosen. We calculated the state transition probability matrix based on the simulated wave

motion data and conducted prediction tests on arbitrarily selected 5-second wave motion processes. When the wave motion interval H_c was set to 0.05m, the prediction results are shown in Fig. 3; adjusting H_c to 0.01m is displayed in Fig. 4.

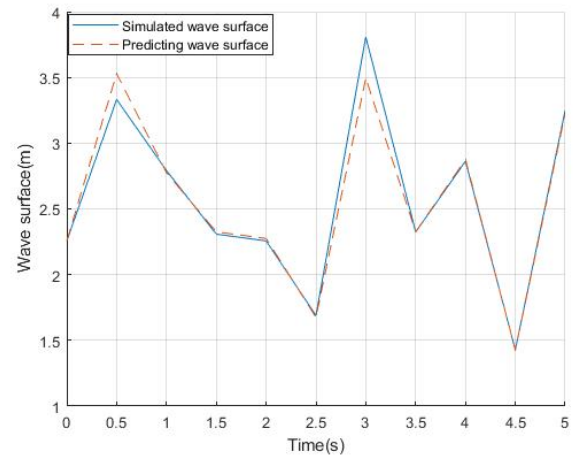


Fig. 3. Prediction results at $H_c = 0.05m$

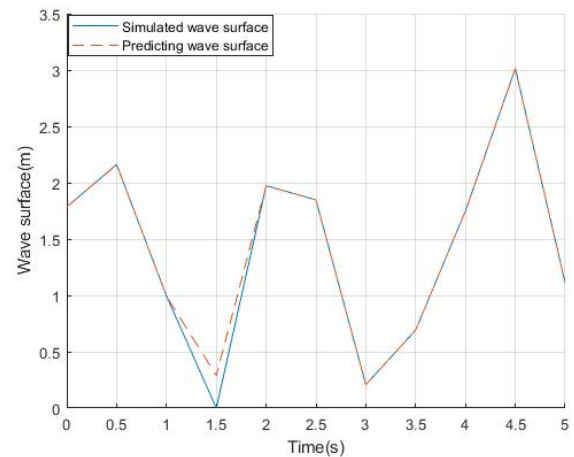


Fig. 4. Prediction results at $H_c = 0.01m$

To objectively assess the accuracy of the prediction model, we employed the Mean Squared Error (MSE) as the measurement criterion, comparing the difference between the predictions and the test set. Tests were conducted under two sea-level change intervals ($H_c = 0.05m$ and $H_c = 0.01m$). The results of the tests indicate that at a smaller sea-level change interval, precisely $H_c = 0.01m$, the accuracy of the model predictions significantly improves. This reflects that more detailed state divisions can more accurately capture the subtle changes in sea-level fluctuations, thereby reducing prediction errors.

The analysis of the results presented in Table 2 shows that when using the Markov chain model for prediction, the accuracy of the predictions can fluctuate significantly due to the reliance on stochastic simulation methods to determine the next state. Nevertheless, it still provides a degree of accuracy.

Table 2. Comparison of MSE

	$H_c = 0.05m$	$H_c = 0.01m$
1	0.4687	0.2684
2	0.2176	0.1604
3	0.4438	0.1113
4	0.1228	0.2027
5	0.5426	0.0336

Currently, most research focuses on predicting long-term changes in sea wave height using neural network methods, with relatively few studies on short-term sea-level fluctuations. Therefore, this paper compares LSTM and BP neural networks with the parameters of the network are shown in Table 3. The mean square error and mean absolute error (MAE) for the models were calculated five times, and their average values are presented in Table 4.

Table 3. Parameters of LSTM and BP

	<i>LSTM</i>	<i>BP</i>
Number of Layers	1	1
Number of Neurons	64	8
Learning Rate	0.005	0.01
Activation function	tanh	tansig

Table 4. Comparison of Models

	<i>MSE</i>	<i>MAE</i>
Our Model	0.3591	0.28652
LSTM	0.98316	0.75069
BP	0.72672	0.57984

The data reveals that, in terms of the accuracy of short-term sea-level change predictions, the Markov chain model outperforms both LSTM and BP neural networks. While LSTM has advantages in time series prediction, the high randomness and lack of transparent time series relationships in short-term sea-level fluctuations lead to poorer results; in contrast, BP neural networks show relatively better accuracy in regression analysis.

Although the Markov chain model can improve the accuracy of sea-level fluctuation predictions to a certain extent, its prediction precision still requires enhancement. The prediction outcomes are influenced by various factors, particularly the choice of dataset, which significantly affects the results. The dataset, primarily derived from numerical simulations, needs to adequately consider the effects of surrounding environmental factors

such as sea winds and currents on sea-level fluctuations, making the state transition process appear more random and thereby affecting the precision of predictions. The data also indicates that when the sea-level change interval is set to $H_c = 0.01m$, the accuracy of predictions is higher because a shorter wave height division interval implies more sea-level states and finer wave height intervals, thereby increasing the probability of random numbers falling within the correct wave height interval.

5 Conclusions

This paper delves into the analysis of sea-level fluctuations, initially conducting a numerical simulation of sea-level movements using Matlab. We conceptualize sea-level fluctuations as a composite of linear waves, determining the amplitude of fluctuations based on the wave spectrum. Subsequently, the study employs a Markov chain model to predict the state of fluctuations. Using data from numerical simulations, we constructed a state transition probability matrix and predicted future fluctuation states through random simulation methods.

Although we successfully utilized the wave spectrum for numerical simulation of sea-level fluctuations, the accuracy of the predictions needed to satisfy expectations. This shortfall is primarily attributed to the poor quality of the data set and limitations in the prediction method. The dataset was generated solely through numerical simulations, resulting in significant deviations from actual sea-level movements. In predicting state transitions, while the random simulation method maintained the stochastic nature of the Markov chain, it also reduced the accuracy of the predictions.

To enhance the accuracy of predictions, it is advisable to consider better state transition methods to reduce the fluctuations in accuracy caused by stochastic simulation methods. Furthermore, exploring the combination of tools capable of handling complex non-linear relationships, such as deep learning, with the Markov chain model to obtain more precise state transition probability matrices could be beneficial. Additionally, considering the interaction between wind, waves, and currents, treating the entire environment as a unified system for state division, and predicting changes in environmental states will further improve the accuracy of predictions.

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